



Commodity prices and fiscal policy in a commodity exporting economy[☆]



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ABSTRACT

This paper analyzes the macroeconomic effects of commodity price shocks on a commodity exporting country. In doing so, we use a DSGE model developed to describe the business cycle in Chile, a copper exporting country. We compare the effects of commodity-price shocks under different fiscal rules. The results show that if the fiscal policy is conducted in a way such that the government saves most of the extra revenues from the higher commodity price, then the macroeconomic effects of a commodity price increase of 10% are an expansion of output below 0.2% and a real exchange appreciation of 0.5%. In contrast, when fiscal policy is highly expansive, the same commodity price increase implies an output expansion above 0.5% and a real exchange rate appreciation of 0.8%. With our model, we also analyze the effects of persistent reduction in the commodity price, the relevance of exchange rate flexibility and the role of imperfect credibility of the fiscal rule.

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1. Introduction

Terms of trade fluctuations are an important source of volatility in most emerging market economies. The recent boom and bust cycle in commodity prices has evidenced their relevance in the economic performance of commodity exporters.¹ In the case of Chile, business cycles have been historically associated to fluctuations in the price of copper, a commodity that represents approximately 40% of total exports and about 15% of public revenues.² Given the importance of copper price movements on the volatility of fiscal revenues, and potentially on output fluctuations, the fiscal authority introduced in

2001 a new fiscal rule to isolate government expenditure from these movements.

In this paper we analyze the effects copper price shocks on different macroeconomic variables from a general equilibrium perspective, with particular emphasis on the role played by the fiscal rule currently in place in Chile. There are several potential mechanisms through which a shock to the copper price may affect the business cycle and variables such as the real exchange rate and inflation. To encompass different first and second round effects, we use a dynamic stochastic general equilibrium (DSGE) model tailored to the Chilean economy (Medina and Soto, 2007). We analyze the impulse response functions to a copper price shock, comparing the results under different fiscal rules. In one case, the government consumes all the extra revenues from the higher copper price. In a second case, we assume the government transfers the extra revenues to the domestic private sector by reducing taxes. Finally, we consider the effects under the *structural balance fiscal rule* introduced in Chile in 2001. This fiscal rule, based on a target for the structural balance as percentage of GDP, is an attempt to compromising and signaling the fiscal policy over a medium-term horizon. The rule allows automatic stabilizers to work uninhibited, while avoiding fine-tuning the fiscal policy to the phases of the cycle.

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¹ See IMF (2012) and IMF (2015) for a cross-country analysis of the effects of commodity price swings in commodity exporters.

² See Spilimbergo (2002) and Caballero (2001), among others.

We consider that a fraction of households do not have access to the capital market. These households consume their disposable income period by period (non-Ricardian households).³

The results show that when the fiscal policy is highly expansive in response to a transitory shock, and if an important fraction of households are non-Ricardian, then an increase in the copper price of 10% implies an output expansion above 0.5%. The real exchange rate appreciates about 0.8% and inflation slightly falls. The size of the GDP expansion is similar to the one found in other studies. For instance, Drexler et al. (2001) report that an increase in the copper price by 10% increases Chile's non-mineral GDP in about 0.5%. In a more recent work, Schmitt-Grohé and Uribe (2015) estimate country-specific SVARs using data from 38 poor and emerging countries. They find that a terms of trade shock of 10% causes a median increase of 0.4% in GDP.

In contrast, if the fiscal policy is conducted in a way such that the government saves most of the extra revenues from the higher copper price (such as the *structural balance* rule), then output would increase between 0.1 and 0.2%. The real appreciation of the exchange rate would also be moderated around 0.5%. Our simulations also show that this type of fiscal policy reduces the adjustment costs of the economy after a persistent decline in copper price.

We show that the adoption of a flexible exchange rate regime also contributes to isolate output and the real exchange rate from copper price shocks. Therefore, the adoption of both the *structural balance* fiscal rule and a flexible exchange rate regime have contributed to reduce the sensitivity of the Chilean GDP to copper price shocks. Although the Chilean economy is not fully immune to copper price fluctuations, these two policies have helped to reduce the vulnerability of the economy to this type of fluctuations.

Our analysis also highlights that imperfect credibility about the *structural balance* fiscal policy would tend to amplify the effects of the copper price shocks on GDP, consumption, and the real exchange rate. However, investment would respond by less under imperfect credibility. Since investment has inertia, it depends heavily on the expected path of the interest rate in the future. Thus, it increases by less in response to a copper price increase due to an expected tighter path for the monetary policy stance under imperfect credibility regarding fiscal policy.

The rest of the paper is organized as follows. The next section reviews related literature. The third section summarizes the main recent developments regarding the fiscal policy in Chile. The fourth section describes the model. Then, the fifth section discusses the solution and parametrization of the model. The sixth section presents and discusses the impulse-response function of shocks to the copper price under different regimes for fiscal and monetary policies. Finally, section seventh concludes.

2. Literature review

Commodity exporting countries have typically experienced more volatile business cycles than other countries due, in part, to the inherent volatility of commodity prices. Several models and estimation approaches have been proposed to understand the relevance of commodity prices in the business cycles of small open economies. Many of these economies tend to have procyclical macroeconomic policies, that exacerbate the sensitivity of the business cycles to commodity price fluctuations.⁴

³ Alternatively, we could think of different behavioral rules for different households. For example, we could just assume that a fraction of households do not intertemporally optimize and just consume their disposable income ("rule of thumb" behavior for these agents). Nonetheless, the results would be identical.

⁴ See Frankel et al. (2013) for evidence on the procyclicality of fiscal policy and Vegh and Vuletin (2013) for the procyclicality of monetary policy.

Pieschacón (2012), analyzing Mexico and Norway, argues that business cycle vulnerability to oil price shocks is related to fiscal policy. Mexico has a higher exposure to oil price fluctuations due to a procyclical fiscal policy while in Norway fiscal policy has been designed to insulate business cycles from oil price shocks. García et al. (2011) have arrived to similar conclusions. They also find that an acyclical fiscal rule unambiguously benefits credit-constrained consumers, but reduces welfare of non-constrained consumers. A recent work for Colombia (Ojeda et al., 2016) illustrates the potential benefits of a countercyclical fiscal rule in an emerging economy that exports oil. Kumhof and Laxton (2010) also suggest the significance of fiscal policy to attenuate the sensitivity of an economy to commodity price fluctuations.

Our work is also related to the research on how commodity price booms generate Dutch Disease effects and how policies can deal with them. For instance, Van Wijnbergen (1984), Krugman (1987), Caballero and Lorenzoni (2014), Lama and Medina (2012), and García-Cicco and Kawamura (2015) evaluate alternative policy responses and instruments in the context of Dutch disease episodes. Although it is possible to consider different instruments to deal with commodity price fluctuations such as exchange interventions, capital control and macro-prudential regulations, our focus in this work is mainly on fiscal policy.

Despite developing economies have improved the conduct of fiscal and monetary policies as argued by Céspedes and Velasco (2014), Frankel et al. (2013), and Vegh and Vuletin (2013), some concerns prevail regarding how to best manage commodity boom-bust cycles in these countries.⁵ In this respect, the case of Chile is interesting because it changed its fiscal and monetary arrangements at the beginning of 2000s with the purpose of achieving better macroeconomic stability. In 1999, Chile adopts a full-fledge inflation targeting regime with a fully flexible exchange rate and, in 2001, it adopts a *structural balance* fiscal rule, making government spending a function of long-term levels of GDP and copper price.

Our major contribution is to make a quantitative analysis in the case of Chile, where a significant fraction of the GDP and fiscal revenues are related to the copper sector. Our model is built to reflect the features of the Chilean economy, the particular role of the copper sector and the specific form for the fiscal and monetary policies in place since 2000.

Finally, it is worth noting that the model developed below is a medium scale model for a small open economy with nominal and real rigidities similar to Adolfson et al. (2007), Altig et al. (2011), Christiano et al. (2005), and Smets and Wouters (2007). We extend this type of models to include a commodity sector that is an endowment that only generates income effects in the economy. Thus, investment in the commodity sector is not considered as in Fornero and Kirchner (2014), Pieschacón (2012), and Ojeda et al. (2016). This is clearly a simplification, but it allows us to quantify the amplifying role of a procyclical fiscal policy by comparing the aggregate effects of a copper price shock when the government spends all current fiscal revenues and when it follows the *structural balance* rule designed in Chile. At the same time, the model includes other complexities as price rigidities, which allow us to analyze the interaction between fiscal and monetary policies.

3. Recent developments in the fiscal rule of Chile

As mentioned above, since 2001 the Chilean government has conducted its fiscal policy through a fiscal rule based on a measure of the Government's *structural balance*. Importantly, this policy was

⁵ See Adler and Magud (2015) and IMF (2015).

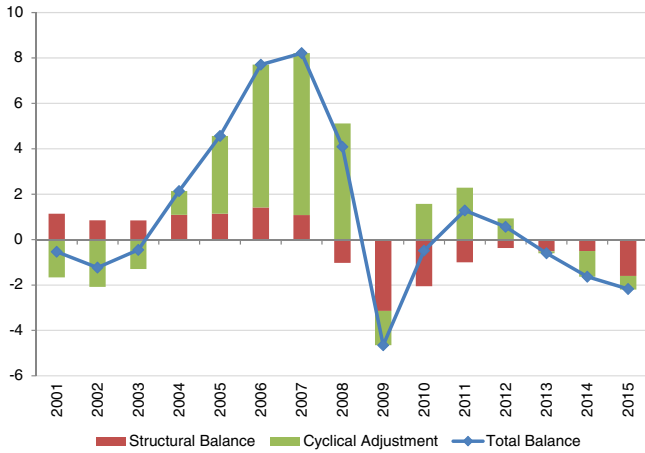


Fig. 1. Fiscal balance in Chile as percentage of GDP.

formalized in 2006 through the Fiscal Responsibility Law. As in the model below, the structural balance is computed in practice by subtracting the cyclical component of revenues to the total balance. In the case of Chile, there are two main sources of cyclical fluctuation in revenues: fluctuations of GDP around its potential and fluctuation of the copper price around its long-run value. The precise methodology on how to adjust for these cyclical components and how to treat specific revenues has been amended along the years. The two key parameters to compute the structural balance, potential output and long-run price of copper, are defined by two experts committees autonomous from the government.

The fiscal rule works as follows: year by year the government sets a target for its structural balance the next year. Then it produces a forecast of its revenues and, based on the structural parameters, computes the cyclically adjusted –or structural– revenues. Finally, it defines its government spending in order to meet (ex-ante) the target for the *structural balance*. Initially, between 2001 and 2007, the target was a structural surplus of 1% of GDP. The rationale for this target was the need to accumulate assets in order to meet different contingent liabilities of the public sector.

After a report on fiscal sustainability (see Velasco et al., 2007), the target for 2008 was reduced to 0.5% of GDP. Later, as a way to generate a countercyclical fiscal policy in response to the global financial crisis, the target was further reduced to 0% of GDP.

In 2010 a special commission revised the methodology and suggested a series of modifications in the way the structural revenues are computed, and to make more transparent the procedures to compute cyclical components of the fiscal revenues (Corbo et al., 2011). After these modifications, the structural balance measured ex-post reached –3.1% of GDP in 2009. Instead of establishing yearly targets, the compromise of the government at the time was to reach a structural balance of –1% of GDP by 2014 (see Fig. 1). In the recent years, the copper price and GDP growth have declined markedly. These situations have implied a sharp deterioration in the structural fiscal revenues of Chile. As a consequence, in 2015 the government assumed as a compromise, a gradual reduction in the structural deficit of 0.25% of GDP per year. Nevertheless, the fiscal rule still allows the automatic fiscal stabilizers to operate: the structural deficit reduction is small, whereas the predicted effective deficit is expected to rise from 2015 to 2016. This reflects that the *structural balance* fiscal rule in Chile has provided fiscal discipline, with enough fiscal buffers to implement a gradual adjustment to persistent lower revenues.

4. The model

As mentioned above, the model is based on Adolfson et al. (2007), Altig et al. (2011), Christiano et al. (2005), and Smets and Wouters (2007). The model has two domestic sectors. Firms in one sector produces differentiated goods that are consumed domestically and exported abroad. Another sector produces a commodity good (copper) that is completely exported abroad. Production in this sector requires no input and it is just a stochastic endowment of copper.⁶ A share of revenues from copper production is owned by the government; the rest is owned by foreign investors. Firms producing differentiated domestic goods use a constant return technology with two inputs, capital and labor. In the steady state, the economy grows at a rate g_y . Consumption exhibits habit formation and there are adjustment cost for investment. Monetary policy is conducted through a policy rule for the interest rate. The model is parameterized using some standard values in the literature, and others are chosen in order to match some steady state ratios for Chile.

4.1. Households

The domestic economy is inhabited by a continuum of households indexed by $j \in [0, 1]$. The expected present value of the utility of household j is given by

$$U_t(j) = E_t \left\{ \sum_{i=0}^{\infty} \beta^i \left(\frac{[C_{t+i}(j) - h(1 + g_y)H_{t+i}]^{\frac{\sigma_C - 1}{\sigma_C}}}{1 - 1/\sigma_C} - \zeta_{L,t} \frac{l_{t+i}(j)^{1 + \sigma_L}}{1 + \sigma_L} + \frac{\zeta_{\mathcal{M}}}{\mu} \left(\frac{\mathcal{M}_{t+i}(j)}{P_{C,t+i}} \right)^{\mu} \right) \right\}, \quad (1)$$

where $l_t(j)$ is labor effort, $C_t(j)$ is a consumption bundle and $\mathcal{M}_t(j)$ corresponds to nominal balances held at the beginning of period t by household j . Parameters σ_C and σ_L are the intertemporal elasticity of substitution for consumption and the inverse elasticity of labor supply with respect to real wages, respectively.⁷ Preferences display habit formation, which are measured by parameter h . The external habit is defined as $H_t = C_{t-1}$, where C_t is the aggregate per capita consumption in period t . All variables are expressed as average values over the members of the household or in per capita terms.

The consumption bundle is a CES aggregator that includes domestically produced goods (home goods) and imported goods (foreign goods):

$$C_t(j) = \left[\gamma_C^{\frac{1}{\eta_C}} C_{H,t}(j)^{\frac{\eta_C - 1}{\eta_C}} + (1 - \gamma_C)^{\frac{1}{\eta_C}} C_{F,t}(j)^{\frac{\eta_C - 1}{\eta_C}} \right]^{\frac{\eta_C}{\eta_C - 1}}, \quad (2)$$

where $C_H(j)$ and $C_F(j)$ are the domestic and imported goods consumed by household j , respectively. Parameter γ_C defines the share of domestic goods in the consumption basket and η_C is the elasticity of substitution between domestic and foreign consumption goods. For any level of consumption, each household purchases a composite of domestic and imported goods in period t in order to minimize the total cost of its consumption basket. Hence, each household minimizes $P_{H,t} C_{H,t}(j) + P_{F,t} C_{F,t}(j)$, subject to (2), where $P_{H,t}$ and $P_{F,t}$ are the prices of domestic and imported goods sold domestically, respectively.

⁶ Copper production is intensive in capital, but a large share of the value added corresponds to rents associated to natural resources.

⁷ To ensure that per capita hours are stationarity, we assume that $\zeta_{L,t}$ grows at a rate $(1 + g_y)^{1 - \frac{1}{\sigma_C}} - 1$.

4.1.1. Budget constraint and consumption-savings decisions

We assume there are two types of households: Ricardian and non-Ricardian households (see Galí et al., 2007). Non-Ricardian households are indexed in the interval $[0, \lambda]$. In other words, λ corresponds to the share of non-Ricardian households in the economy. These households have no access to the capital market and, therefore, can not smooth consumption intertemporally.

4.1.1.1. Ricardian households. These households have access to three different types of assets: money $\mathcal{M}_t(j)$, one-period non-contingent foreign bonds (denominated in foreign currency) $B_{p,t}^*(j)$, and one-period domestic contingent bonds $d_{t+1}(j)$ which pays out one unit of domestic currency in a particular state. There are no adjustment costs in the portfolio composition of domestic assets. However, each time a domestic household borrows from abroad it must pay a premium over the international price of external bonds. This premium is introduced in the model to obtain a well defined steady state for the economy.⁸ Hence, the household budget constraint is given by

$$P_{C,t}C_t(j) + E_t[q_{t,t+1}d_{t+1}(j)] + \frac{\varepsilon_t B_{p,t}^*(j)}{(1 + i_t^*)\Theta\left(\frac{\varepsilon_t B_t^*}{P_{Y,t}Y_t}\right)} + \mathcal{M}_t(j) = W_t(j)l_t(j) + \Pi_t(j) - T_{p,t}(j) + d_t(j) + \varepsilon_t B_{p,t-1}^*(j) + \mathcal{M}_{t-1}(j), \quad (3)$$

where $\Pi_t(j)$ are profits received from domestic firms, $W_t(j)$ is the nominal wage set by household j , $T_{p,t}(j)$ are per capita net taxes, and ε_t is the nominal exchange rate. The term $\Theta(\cdot)$ corresponds to the premium that domestic households have to pay each time they borrow from abroad which depends on the ratio of net foreign asset position of the country to GDP, where B_t^* is the aggregate net foreign asset position of the economy and $P_{Y,t}Y_t$ is the nominal GDP. Variable $q_{t,t+1}$ is the period t price of domestic contingent bonds normalized by the probability of the occurrence of a particular state.

Ricardian households choose consumption and the composition of their portfolios by maximizing Eq. (1) subject to Eq. (3). Since we are assuming the existence of a complete set of contingent claims, consumption is equalized across Ricardian households.

4.1.1.2. Non-Ricardian households. As we said, these households have no access to the capital market and own no share in domestic firms. Therefore, they must consume completely their disposable labor income, period by period.⁹

$$P_{C,t}C_t(j) = W_t(j)l_t(j) - T_{p,t}(j). \quad (4)$$

where $j \in [0, \lambda]$.

4.1.2. Labor supply and wage setting

Each household j is a monopolistic supplier of a differentiated labor service. There is a set of perfectly competitive labor-service assemblers that hire labor from each household and combine it into an aggregate labor service unit, l_t , that is then used by the intermediate goods producer. The labor service unit is defined as

$$l_t = \left(\int_0^1 l_t(j)^{\frac{\varepsilon_L-1}{\varepsilon_L}} dj \right)^{\frac{\varepsilon_L}{\varepsilon_L-1}}. \quad (5)$$

where $l_t(j)$ corresponds to the labor supply of household j and ε_L is the elasticity of substitution of the household labor supply. The

optimal composition of this labor service unit is obtained by minimizing its cost, given the different wages set by different households. In particular, the demand for the labor service provided by household j is

$$l_t(j) = \left(\frac{W_t(j)}{W_t} \right)^{-\varepsilon_L} l_t, \quad (6)$$

where $W_t(j)$ is the wage rate set by household j and W_t is an aggregate wage index defined as $W_t = \left(\int_0^1 W_t(j)^{1-\varepsilon_L} dj \right)^{\frac{1}{1-\varepsilon_L}}$.

Following Erceg et al. (2000), we assume that wage setting is subject to a nominal rigidity à la Calvo (1983). In each period, each type of household faces a constant probability $(1 - \phi_L)$ of being able to re-optimize its nominal wage. We assume there is an “passive” updating rule for all those households that cannot re-optimize their wages. This “passive” updating rule states that workers who are unable to optimally reset their wages, update them by considering a geometric weighted average of past CPI inflation and the inflation target set by the authority, $\bar{\pi}_t$, where ξ_L is the weight to past inflation. This rule also includes the productivity growth rate in order to avoid a divergent real wage dispersion along the steady state growth path. Once a household has decided a wage, it must supply any quantity of labor service that is demanded at that wage.

For simplicity we assume that non-Ricardian households set wages equal to the average wage set by Ricardian households. Given the labor demand for each type of labor, this assumption implies that labor effort of non-Ricardian households coincides with the average labor effort by Ricardian households.

4.2. Investment and capital goods

Investment goods consist of a CES aggregator of home ($I_{H,t}$) and foreign goods ($I_{F,t}$):

$$I_t = \left[\gamma_I^{\frac{1}{\eta_I}} I_{H,t}^{\frac{\eta_I-1}{\eta_I}} + (1 - \gamma_I)^{\frac{1}{\eta_I}} I_{F,t}^{\frac{\eta_I-1}{\eta_I}} \right]^{\frac{\eta_I}{\eta_I-1}},$$

where η_I is the elasticity of substitution between home and foreign investment goods, and γ_I defines the share of domestic goods in investment. Maximizing the expression above subject to the budget constraint $P_{I,t}I_t = P_{H,t}I_{H,t} + P_{F,t}I_{F,t}$ we obtain the investment demand functions for $I_{H,t}$ and $I_{F,t}$.

To capture the observed inertia in the demand for investment goods, we assume that adjusting investment is costly. A representative firm chooses a path for investment that maximizes the present value of its profits:

$$\max_{K_{t+i}, I_{t+i}} E_t \left\{ \sum_{i=0}^{\infty} \Lambda_{t,t+i} \frac{[Z_{t+i}K_{t+i} - P_{I,t+i}I_{t+i}]}{P_{C,t+i}} \right\},$$

subject to

$$K_{t+1} = (1 - \delta)K_t + S\left(\frac{I_t}{I_{t-1}}\right)I_t.$$

where K_t is the amount of physical capital available at the beginning of period t , and Z_t is the rental rate per unit of effective capital. Function $S(\cdot)$ characterizes the adjustment cost for investment.¹⁰

⁸ See Schmitt-Grohé and Uribe (2003) for different ways to get a steady state independent of initial conditions for small open economy models.

⁹ For simplicity we assume that non-Ricardian households use money for transactions, but not for savings.

¹⁰ The adjustment cost of investment satisfies the following: $S((1 + g_y)) = 1, S'((1 + g_y)) = 0, S''((1 + g_y)) = -\mu_S < 0$ (see Altig et al., 2011).

4.3. Domestic production

This sector consists of two types of firms. One type of firms are producers of differentiated intermediate goods. Each of these firms has monopoly power and face a nominal rigidity that prevents them to adjust optimally prices every period. A second type of firms assemble the differentiated intermediate goods to sell them in the domestic and foreign markets. This last type of firms behave competitively.

4.3.1. Assembly of intermediate goods

Assemblers of intermediate goods sell different final goods in the domestic and the foreign market. In order to produce $Y_{H_D,t}$ units of home goods to be sold in the domestic market, they combine domestically produced intermediate varieties by using the following aggregator:

$$Y_{H_D,t} = \left[\int_0^1 Y_{H_D,t}(z_H)^{\frac{\epsilon_{H_D}-1}{\epsilon_{H_D}}} dz_H \right]^{\frac{\epsilon_{H_D}}{\epsilon_{H_D}-1}}, \quad (7)$$

where $Y_{H_D,t}(z_H)$ is the quantity of intermediate variety z_H used for final good sold in the domestic market and ϵ_{H_D} is the elasticity of substitution among varieties.

Analogously, in order to export $Y_{H_F,t}$ units of the home good, assemblers combine intermediate varieties using the following technology:

$$Y_{H_F,t} = \left[\int_0^1 Y_{H_F,t}(z_H)^{\frac{\epsilon_{H_F}-1}{\epsilon_{H_F}}} dz_H \right]^{\frac{\epsilon_{H_F}}{\epsilon_{H_F}-1}}, \quad (8)$$

where $Y_{H_F,t}(z_H)$ is the amount of variety z_H used to export $Y_{H_F,t}$ units of the final home good and ϵ_{H_F} is the elasticity of substitution among varieties.

4.3.2. Producers of intermediate goods

The production of intermediate goods is characterized by firms that act as a monopoly in the production of a single variety. These firms differentiate their production depending on whether is used by exporting assemblers or by domestic market assemblers. Each firm maximizes profits by choosing the price of its variety subject to the corresponding demand and the available technology. Let $Y_{H,t}(z_H) = Y_{H_D,t}(z_H) + Y_{H_F,t}(z_H)$ be the total quantity produced of a particular variety z_H , where $Y_{H_D,t}(z_H)$ corresponds to the quantity produced for the domestic market and $Y_{H_F,t}(z_H)$ the quantity produced for the foreign (exports) market. The technology available is given by

$$Y_{H,t}(z_H) = A_{H,t} \left[\eta_H^{\frac{1}{\theta_H}} ((1+g_y)^t l_t(z_H))^{\frac{\theta_H-1}{\theta_H}} + (1-\eta_H)^{\frac{1}{\theta_H}} (K_t(z_H))^{\frac{\theta_H-1}{\theta_H}} \right]^{\frac{\theta_H}{\theta_H-1}},$$

where $l_t(z_H)$ is the per capita amount of labor used, $K_t(z_H)$ is the amount of physical capital rented. Variable $A_{H,t}$ represents a productivity shock common to all firms in this sector. Parameter θ_H is the elasticity of substitution between labor and capital services and η_H controls the share of labor services in production. Firms determine their optimal mix of inputs by minimizing the total cost of production, subject to the constraint imposed by the technology.

Following Calvo (1983) we assume that firms adjust their prices infrequently. In particular, they do so when receiving a signal. In every period the probability of receiving a signal and adjusting their prices in the domestic market is $1 - \phi_{H_D}$ for all firms, independently of their history. Similarly, each firm has a probability receiving a signal to be able to re-optimize its exporting price equal to $1 - \phi_{H_F}$. The

chance of receiving this other signal is equal for all firms, and independent of their history and of the event of adjusting optimally prices in the domestic market. We assume that a firm that does not receive any type of signal will update its prices following a simple “passive” updating rules based on a weighting average between past inflation and the inflation target.

4.4. Import goods retailers

The import sector consists of a continuum of firms that buy a homogenous good in the foreign market. These firms transform the imported good in a continuum of differentiated imported inputs.¹¹ Competitive assemblers combine this continuum of differentiated imported inputs in a final import good Y_F . The technology of importing assemblers is given by

$$Y_{F,t} = \left[\int_0^1 Y_{F,t}(z_F)^{\frac{\epsilon_F-1}{\epsilon_F}} dz_F \right]^{\frac{\epsilon_F}{\epsilon_F-1}}, \quad (9)$$

where $Y_{F,t}(z_F)$ is the quantity of a differentiated import z_F used by the assemblers and ϵ_F is the elasticity of substitution among differentiated imported goods.

The different importing firms buy abroad the homogenous foreign good at price $P_{F,t}^*$ in foreign currency. Each different importing firm has monopoly power over the domestic retailing of the specific variety. We assume local currency price stickiness in order to allow for an incomplete exchange rate pass-through into the domestic prices. An importing firm adjusts the domestic price of its variety infrequently, when receiving a signal. The signal arrives with probability $1 - \phi_F$ each period. As in the case of domestically produced goods, if a firm does not receive a signal, it updates its price following a “passive” rule based on a weighting average between past inflation and the inflation target.

4.5. Commodity and foreign sectors

We assume that the production of the commodity requires no inputs: there is an exogenous endowment of the commodity good. This endowment is completely exported and it can be interpreted as the value added by natural resources in the commodity gross production. The endowment of the commodity good grows at rate g_y to guarantee a sectoral balanced growth.

Foreign agents demand the commodity good, and domestic goods assembled by the intermediaries. The demand for the commodity good is completely elastic at the price $P_{S,t}^*$. The law of one price holds for this good. Therefore, the domestic currency price of the commodity is given by

$$P_{S,t} = \varepsilon_t P_{S,t}^*, \quad (10)$$

We assumed that the log-deviation of the real price copper (price of copper deflated by the foreign price level) can be represented as an AR(1) process,

$$\hat{p}r_{S,t}^* = \rho_{cu} \hat{p}r_{S,t-1}^* + \varepsilon_{cu,t} \quad (11)$$

where $0 < \rho_{cu} < 1$ and $E_{t-1} \varepsilon_{cu,t} = 0$.

¹¹ This differentiating technology can be interpreted as brand naming.

The real exchange rate is defined as the relative price of a foreign price level and the price of a consumption basket in the domestic economy:

$$RER_t = \frac{\mathcal{E}_t P_t^*}{P_{C,t}}. \quad (12)$$

Foreign demand for domestically produced goods depends on the relative price of this type of goods and the total foreign aggregate demand, C_t^*

$$Y_{H_f,t} = \zeta^* \left(\frac{P_{H_f,t}^*}{P_t^*} \right)^{-\eta^*} C_t^*, \quad (13)$$

where ζ^* corresponds to the share of domestic intermediate goods in the consumption basket of foreign agents, and where η^* is the price elasticity of the demand.

4.6. Monetary policy

Monetary policy is defined through a rule for the interest rate on short-term bonds. The rule implies that interest rate adjusts in response to deviations of consumption goods inflation from the inflation target and GDP deviations from its balanced growth path, $\bar{Y}_t$. We also allow for interest rate smoothing:

$$\frac{1+i_t}{1+i} = \left(\frac{1+i_{t-1}}{1+i} \right)^{\varphi_i} \left(\frac{P_{C,t}}{P_{C,t-1}} \frac{1}{1+\bar{\pi}} \right)^{(1-\varphi_i)\varphi_\pi} \left(\frac{Y_t}{\bar{Y}_t} \right)^{(1-\varphi_i)\varphi_y}, \quad (14)$$

where φ_i defines the interest rate smoothing, φ_π and φ_y are the weights of inflation and GDP deviations in the monetary policy rule.

4.7. Fiscal policy

Let us consider the budget constraint of the government. The net position of the government measured in foreign currency, $B_{G,t}^*$, evolves according to

$$\frac{\mathcal{E}_t B_{G,t}^*}{(1+i_t^*) \Theta \left(\frac{\mathcal{E}_t B_{G,t}^*}{P_{Y,t} Y_t} \right)} = \mathcal{E}_t B_{G,t-1}^* + T_t - P_{G,t} G_t,$$

where $(1+i_t^*) \Theta(\cdot)$ is the relevant gross interest rate for government debt, G_t is government expenditure and T_t are total net fiscal nominal revenues (fiscal revenues minus transfers to the private sector). By simplicity, we assume that the basket consumed by the government is only in domestically produced goods, implying that the price deflator of government consumption is $P_{G,t} = P_{H_D,t}$.

Fiscal net revenues come from two sources: income tax to the private sector (net of fiscal transfers), which is a function of GDP, $T_{P,t} = \tau_t P_{Y,t} Y_t$; and revenues from copper which are given by $P_{S,t} \chi Y_{S,t}$. Parameter χ defines the share of the fiscal revenues in total copper production and variable τ_t corresponds to the average net income tax.¹²

The fiscal policy is defined by the three variables: $B_{G,t}$, τ_t , and G_t . Therefore, given the budget constraint of the government, it is necessary to define behavior rules for two of these three variables.

When all agents are Ricardian, the trajectory for the primary deficit is irrelevant for the households decisions, as long as the budget constraint of the government is satisfied. On the contrary, when a fraction of the agents are non-Ricardian then, the precise trajectory of the government debt and the primary deficit are

relevant. Additionally, the path of the government expenditure may be relevant on its own as long as its composition differs from the composition of private consumption.

We consider the following alternative rules for the fiscal policy:

Rule A: Government expenditure adjustment

Under this rule we assume that government expenditure adjusts in order to satisfy budget constraint. Net taxes and the net position of the government are kept constant in response to a shock:

$$B_{G,t} = \text{const}; \quad \tau_t = \text{const} \quad (15)$$

Under this rule, an increase in the price of copper induces automatically an expansion of the government expenditure.

Rule B: Tax and transfers adjustment

Under this rule, net taxes (taxes and or transfers) are adjusted in order to satisfy the budget constraint. Government expenditure (as a share of GDP, in nominal terms) and the net position of the government are kept constant:

$$B_{G,t} = \text{const}; \quad \frac{P_{G,t} G_t}{P_{Y,t} Y_t} = \text{const} \quad (16)$$

Under this rule an increase in the copper price induces a reduction in taxes in order to satisfy the budget constraint

Rule C: Structural balance fiscal rule

The *structural balance* fiscal rule is a rule for the fiscal policy that has been in place in Chile since 2001.¹³ The explicit objective for this rule is to smooth the path of government expenditure in order to avoid a procyclicality in the fiscal policy. As it will be clear, this rule allows for a change in the net asset position of the government together with an endogenous adjustment in government expenditure and/or net taxes.

Consider the balance of the government,

$$BA_t = T_t - P_{G,t} G_t + \left(1 - \frac{1}{(1+i_{t-1}^*) \Theta_{t-1}} \right) \mathcal{E}_t B_{G,t-1}^*, \quad (17)$$

Eq. (17) implies that the balance of the government includes interest payments (the last term on the RHS). The structural balance, $B_{S,t}$, is defined as the effective balance minus cyclical revenues:

$$B_{S,t} \equiv BA_t - \tilde{T}_t = T_t - \tilde{T}_t - P_{G,t} G_t + \left(1 - \frac{1}{(1+i_{t-1}^*) \Theta_{t-1}} \right) \mathcal{E}_t B_{G,t-1}^* \quad (18)$$

where $\tilde{T}_t = \tilde{T}_{P,t} + \tilde{T}_{Cu,t} + \varepsilon_{T,t}$ corresponds to cyclical revenues, obtained when the GDP is in its potential level and the copper price is in its long-run value plus an unpredicted shock uncorrelated with copper price and GDP ($\varepsilon_{T,t}$). Formally, the first two components of the cyclical revenues are defined as

$$\tilde{T}_{P,t} = \tau_t P_{Y,t} [Y_t - \bar{Y}_t] \quad \tilde{T}_{Cu,t} = [P_{S,t} - \mathcal{E}_t \bar{P}_{S,t}^{ref}] \chi Y_{S,t} \quad (19)$$

where $\bar{P}_{S,t}^{ref}$ is a long run price of copper, which is called the *reference price*, and $\bar{Y}_t$ is potential output or balanced growth path of GDP.

¹² Seigniorage from the Central Bank is rebated to the private sector as a lump-sum transfer.

¹³ The description of the structural balance fiscal rule is an adaptation of the description of the rule in Marcel et al. (2001).

According to the *structural balance* fiscal rule, the objective of the fiscal policy is to keep the structural balance close to zero.¹⁴ We assume that the net average income tax (τ_t) is kept constant at τ . This assumption implies that our interpretation of the *structural balance* rule may result in a more expansive policy rule than the actual one. In practice, an important share of the increase in government expenditure when the copper reference price rises corresponds to transfers to the private sector. In our model this is equivalent to reducing τ .¹⁵ Combining Eqs. (17), (18) and (19) we obtain an expression for government expenditure, as a share of GDP, that is consistent with the rule C,

$$\frac{P_{G,t}G_t}{P_{Y,t}Y_t} = \left(1 - \frac{1}{(1 + i_{t-1}^*)\Theta_{t-1}}\right) \frac{\varepsilon_t}{\varepsilon_{t-1}} \frac{\varepsilon_{t-1}B_{t-1}^*}{P_{Y,t-1}Y_{t-1}} \frac{P_{Y,t-1}Y_{t-1}}{P_{Y,t}Y_t} + \tau \left(\frac{\bar{Y}_t}{Y_t}\right) + \varepsilon_t \bar{P}_{S,t}^{\text{ref}} \chi \frac{Y_{S,t}}{P_{Y,t}Y_t} - \varepsilon_{T,t} - \frac{B_{S,t}}{P_{Y,t}Y_t} \quad (20)$$

where $\frac{B_{S,t}}{P_{Y,t}Y_t} = \bar{b}_S$ is the target for the structural balance.

It is important to recall that rule C is different than rule B because we allow for an accumulation or de-accumulation of net assets by the government. However, the effects of a copper price shock under this rule would be the same as under rule B when all households are Ricardian.

5. Parametrization and solution

To solve the model we first solve for the non-stochastic steady-state by using numerical methods. Then we solve the log-linearized decision rules from the behavioral equations of the model. We use the QZ factorization described in Uhlig (1998). Table 1 presents the value chosen for the structural parameters of the model.¹⁶ Many of these parameters were taken directly from the literature and based on a recent estimation of a similar model for Chile, such as Fornero and Kirchner (2014). Some other parameters were chosen so as to match macroeconomic statistics for the Chilean economy in 2001–2015.

We use a quarterly frequency for the model. We calibrate annual productivity growth at 2.5% and the share of the commodity export sector over total GDP to 12%, which is consistent with the average value for the Chilean economy since 2001. Net exports to GDP ratio is calibrated to 2%. The foreign debt as percentage of GDP is calibrated in order to imply a current account to GDP ratio of –1.8%, an average value for Chile since 2001. Government spending is calibrated to 12% of the GDP, a similar value used in Fornero and Kirchner (2014).

The parameters of household's preferences are the elasticities of substitution among the different types of goods, and the share of them in the consumption basket. The intertemporal elasticity of substitution is fixed in 1.0 as estimated by Duncan (2003) for Chile. We use a unitary labor supply elasticity, which is lower than traditional value used in U.S. real business cycle models. However, this value is more in line with microeconomic estimates for Chile.¹⁷ The household discount factor (β) is chosen to imply a steady state real interest rate around 3% (annual basis), which is in line with

Table 1
Baseline parametrization.

Parameter	Value	Description
<i>Household preferences</i>		
β	0.996	Subjective discount factor
σ_C	1.0	Elasticity of intertemporal substitution in consumption
σ_L	1.0	Inverse of the labor supply elasticity
h	0.75	Habit formation coefficient
g_y	2.5% (annual)	Productivity growth rate
λ	0.5	Fraction of non-Ricardian Households
<i>Consumption and investment baskets</i>		
γ_C	0.75	Share of domestic goods in consumption
η_C	1.5	Elasticity of substitution in consumption between domestic and imported goods
γ_I	0.65	Share of domestic goods in investment
η_I	0.5	Elasticity of substitution in investment between domestic and imported goods
<i>Nominal rigidities</i>		
ϕ_L	0.875	Prob adjusting wages
ξ_L	0.7	Wage indexation
ϕ_{H_D}	0.66	Prob adjusting P_{H_D}
ξ_{H_D}	0.5	Domestic goods indexation (home)
ϕ_{H_F}	0.875	Prob adjusting P_{H_F}
ξ_{H_F}	0.5	Domestic goods indexation (abroad)
ϕ_F	0.66	Prob adjusting P_F
ξ_F	0.5	Imported goods indexation
<i>Capital accumulation</i>		
μ_S	2	Investment adjustment cost coefficient
δ	6.0% (annual)	Depreciation rate
<i>Domestic production technology</i>		
η_H	0.66	Labor share in domestic production
θ_H	1.0	Elasticity of substitution between labor and capital
<i>Foreign sector</i>		
NX/Y	2%	Net exports to GDP ratio
η^*	0.5	Price elasticity of foreign demand for domestically produced goods
ρ_{cu}	0.95	Persistence of copper price shocks
Q	0.001	Elasticity of the external supply of debt
CA/GDP	–1.8%	Current account to GDP ratio
Y_S/Y	12%	Copper share in total GDP
<i>Monetary policy</i>		
φ_i	0.85	Interest rate smoothing
φ_π	1.75	Reaction to inflation
φ_y	0.10	Reaction to output
<i>Fiscal sector</i>		
G/Y	12%	Government expenditure to GDP ratio
χ	30%	Share of property of commodity sector holds for the government
τ	9.4%	Average net tax rate
$\bar{b}_S$	0%	Structural balance

estimations for the neutral interest rate in Chile (see Fuentes and Gredig, 2008). The elasticity of substitution between domestic and imported consumption goods is set to 1.5, which is consistent with the estimation in Fornero and Kirchner (2014). The share of domestic produced goods in consumption is 75%, which is consistent with the average fraction of domestic goods in the CPI basket since 2001. We set the habit formation coefficient to $h = 0.75$, which is similar to estimations in Adolfson et al. (2007), Medina and Soto (2007) and Fornero and Kirchner (2014).¹⁸

¹⁸ Since monetary policy is described by a rule for the interest rate, money supply will endogenously adjust to the money demand. The separability between money and consumption implies that the money market equilibrium is no needed to characterize the equilibrium of the rest of variables of the model. Consequently, a calibration for interest rate elasticity of the money demand is not required.

¹⁴ As mentioned above, originally the target for the structural surplus was 1% of GDP. This target was reduced in May 2007 to 0.5%. In 2009, the structural balance was set at 0% of GDP.

¹⁵ However, transfers to the private sector are focused on low income households which tend to have a larger marginal propensity to consume (they are “less” Ricardian). Therefore, whether the government increase their consumption or whether they transfer the extra income to low income households, it would probably generate very similar results.

¹⁶ Medina and Soto (2007) and Fornero and Kirchner (2014) present estimated versions of this model.

¹⁷ See Mizala et al. (1998) and Fornero and Kirchner (2014).

In the commodity export sector, we assume that the participation of the government in the copper production is about 30% ($\chi = 0.3$), consistent with the average fiscal income associated with the copper sector since 2001. The persistence of the copper price shocks are set at $\rho_{cu} = 0.95$, which is approximately the value obtained from a quarterly estimation during 2000–2015. The elasticity of the country risk premium with respect to the net foreign asset position is $\varrho = 0.001$, which is a similar value to the one estimated in [Fornero and Kirchner \(2014\)](#).

Capital goods can be classified broadly in two types: equipments-machines and buildings-infrastructure. In the case of Chile, equipments-machines represents about 40 % of total investment and they are mostly imported goods (see [Fornero and Kirchner, 2014](#)). Given the nature of investment, it plausible to consider a lower degree of substitution between these two types. Consequently, for the production function of the capital goods, we assume that the elasticity of substitution between domestic and imported good is 0.5 and the share of domestically produced goods in investment is 65%.¹⁹

The annual depreciation rate of this type of capital is 6%. The investment adjustment cost is set at $\mu_s = 2$, which is similar to the values found in the model estimations in [Adolfson et al. \(2007\)](#), [Medina and Soto \(2007\)](#), [Smets and Wouters \(2007\)](#), and [Fornero and Kirchner \(2014\)](#). Producers of domestic intermediate goods have access to a constant return technology. We assume that the elasticity of substitution between capital and labor is 1 (Cobb-Douglas production function), and that the share of labor is 66% in production. Finally, the foreign demand for domestic traded goods has an elasticity of 0.5, which is in line with recent aggregate estimation for the export demand in Chile (see [Monfort, 2008](#) and [Fornero and Kirchner, 2014](#)).

Several studies have estimated, using various techniques, the share of non-Ricardian consumers in Chile and in other developed and developing countries. In the case of Chile, [Corbo and Schmidt-Hebbel \(1991\)](#) estimate λ equal to 0.60 for the period 1968–1988; [Schmidt-Hebbel and Servén \(1996\)](#) 0.45 for the period 1963–1991; and [Bandiera et al. \(1999\)](#) 0.55 for the period 1970–1995. More recently [López et al. \(2000\)](#), using a panel of developed and developing countries, found a share of constrained consumers of 0.61 for developing countries. In our case, we assume that the fraction of these types of households is $\lambda = 50\%$, which is the same value considered in [Fornero and Kirchner \(2014\)](#).

For the nominal rigidities, we assume that domestically sold goods have prices that are optimally set every three quarters ($\phi_{H_D} = \phi_F = 0.66$). To obtain some degree of inertia in price inflation of these goods, we consider a partial indexation to past inflation ($\xi_{H_D} = \xi_F = 0.5$). These values are not far from estimations in similar models (see, for instance, [Adolfson et al., 2007](#), [Medina and Soto, 2007](#), and [Fornero and Kirchner, 2014](#)). Following the recent estimations in [Fornero and Kirchner \(2014\)](#), we also assume that wages and export prices of non-commodity goods have a higher degree of nominal rigidities ($\phi_L = \phi_F = 0.875$, $\xi_L = 0.7$, $\xi_{H_F} = 0.5$). The baseline monetary policy is based on the recent estimation in [Fornero and Kirchner \(2014\)](#) ($\varphi_i = 0.85$, $\varphi_\pi = 1.75$, $\varphi_y = 0.10$).²⁰

6. Effects of shocks to the commodity price

In this section we analyze the effects of a copper price shocks on several macroeconomic variables. First, we analyze the responses

to a 10% increase in the copper price under the three fiscal rules described above. Second, we show the macroeconomic responses to a 50% persistent decline in the copper price under the same three fiscal rules. Third, we discuss how a less flexible exchange rate affects the responses of the economy to copper price shocks. Finally, we analyze how imperfect credibility on the fiscal rule can modify the responses of the economy to commodity price shocks.

6.1. Comparing different fiscal rules to a copper price increase

As we said before, the first effect of a rise in the copper price is to increase government revenues. What the government does with the revenues from copper depends on the fiscal policy rule. In turn, the macroeconomic effects of the shock under these different policies depend on the fraction of non-Ricardian households. When all households are Ricardian, the precise path of the government debt does not have an impact on the consumption and savings decisions of households.²¹ Thus, we consider the responses of the variables to the commodity price shock assuming that a significant fraction of households are non-Ricardian ($\lambda = 0.5$). These households do not have access to the capital market to smooth consumption and, therefore, consume all their disposable income.

[Figs. 2 and 3](#) present the responses of the main macroeconomic variables to a 10% rise in the copper price. As we discussed before, rule A is the more expansive of the fiscal policies considered as it implies a direct expansion in the aggregate demand. Thus, when a fraction of households are non-Ricardian the impacts of the shock on output and employment are the largest. As the government spends directly the revenues from copper, the expansion in aggregate demand stimulates domestic production which generates a large increase in employment, around 1%. GDP increases by almost 0.6%. Although real wages do not increase by much, the increase in employment results in a rise in disposable income to the private sector. As a consequence, non-Ricardian households expand their consumption which leads to an increase in total private consumption. This effect is the standard Keynesian multiplier effect in textbooks. The expansion in employment leads to an increase in the marginal product of capital and, therefore, in the demand for this factor. Investment, however, responds slowly to the increase in the demand for capital. The expansion in investment is also offset by a higher monetary policy rate that tries to control the GDP expansion. The real exchange rate appreciates about 0.8% on impact. This last effect dominates in total inflation, which falls despite of the rise in real wages.

As we discussed previously, when a fraction of households are non-Ricardian the path followed by government debt and taxes is relevant to determine private consumption. Under rule B the net position of the government vis-a-vis the private sector remains constant as all revenues are transferred to the private sector through a tax reduction. Private consumption increases as disposable income rises. However, since there is a fraction of households that are non-Ricardian, the expansion in consumption is much larger than under rule A. Ricardian households also increase their consumption but to a lesser extent as they smooth consumption intertemporally. Overall, private consumption rises between 0.3 and 0.4%. The real appreciation of the exchange rate modifies the composition of spending, increasing private consumption of imported goods by more than of domestic goods. However, this effect is smaller than under rule A.

¹⁹ Moreover, [Fornero and Kirchner \(2014\)](#) obtain an estimation for this elasticity around 0.5.

²⁰ The elasticity of substitution for household labor supply, varieties of home goods and differentiated imported goods are not presented in [Table 1](#). We assume a mark-up of 10% in these markets, which implies $\epsilon_L = \epsilon_{H_D} = \epsilon_{H_F} = \epsilon_F = 11$. Nevertheless, the relevance of these parameters is finally captured for the national account shares at the steady state, which have been selected according to the Chilean economy since 2001.

²¹ Responses under the case with only Ricardian households are available upon request.

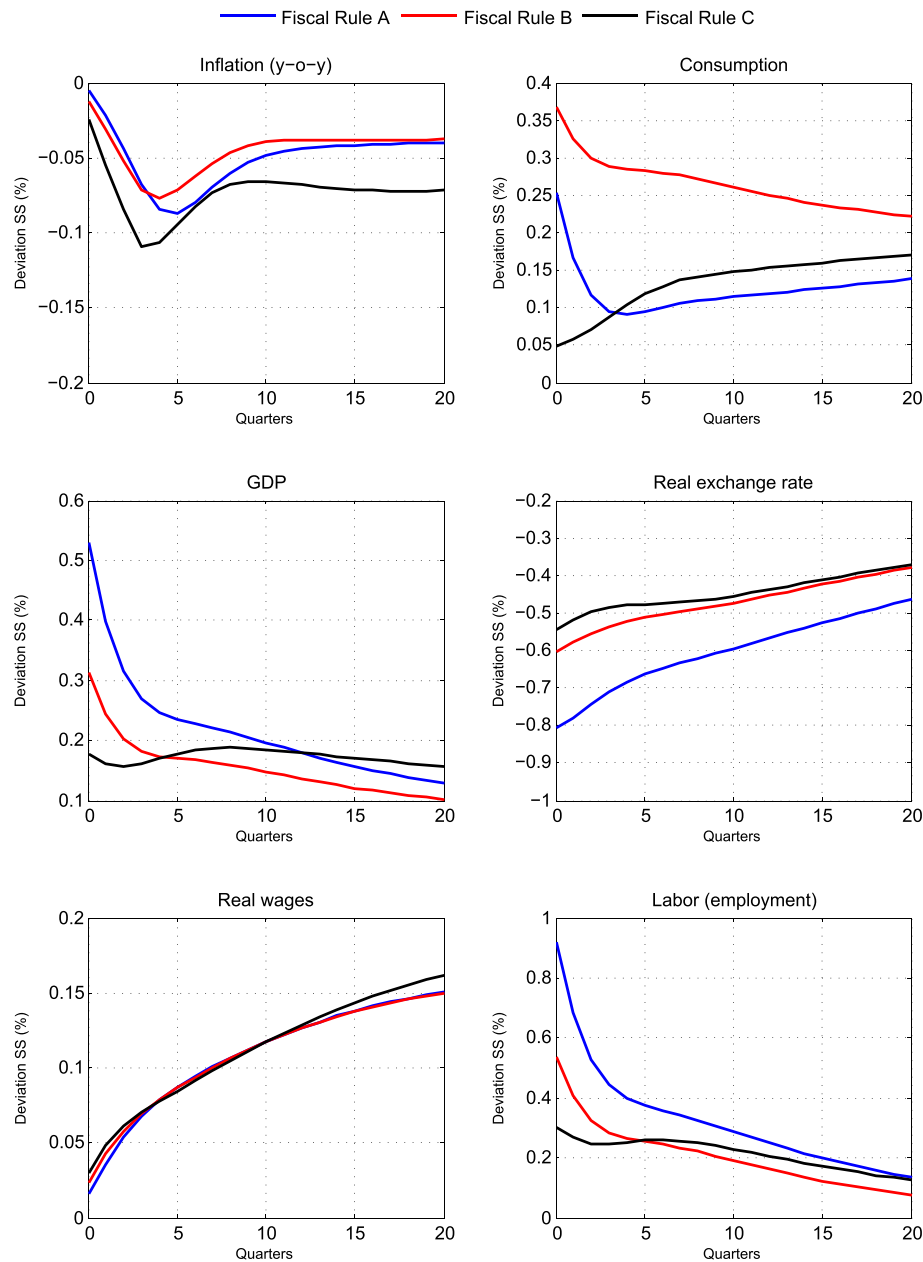


Fig. 2. Impulse-response to a copper price increase.

GDP rises by 0.3% on impact after the shock. This expansion in output reflects in an increase in employment. As before, real wages increase only slightly due to the nominal wage rigidities in the economy, but the real appreciation of the currency of 0.6% leads to a reduction in total inflation.

Let consider now the effects under rule C, when government expenditure is determined by the *structural balance* rule. Consumption by Ricardian households increase immediately after the shock because they know that all asset accumulation by the government will eventually give rise to a reduction in taxes in the future. Non-Ricardian households increase their consumption slowly as their disposable income increase through a smaller rise in employment. The path of private consumption, therefore, is increasing over time, reaching a range between 0.1 and 0.2% three years after the shock. Consistent with the previous result, there is an expansion in

aggregate demand. Despite the fact that foreign demand for domestic goods falls, as the real exchange appreciates, demand for domestic goods rises. Although real wages increases in a similar magnitude as under the other fiscal rules and the real appreciation is smaller, total inflation reduces a little more as consequences of smaller expansion of aggregate demand. The maximum expansion of GDP under this rule is less than 0.2%. The current account, as a share of GDP, improves by 0.4%, which is similar to rules A and B, but with a smaller real exchange appreciation of around 0.5%.

6.2. Comparing the effects of different fiscal rules to a persistent copper price decline

In recent years, commodity prices have declined markedly. Copper price has not been an exception in this recent trends. Thus,

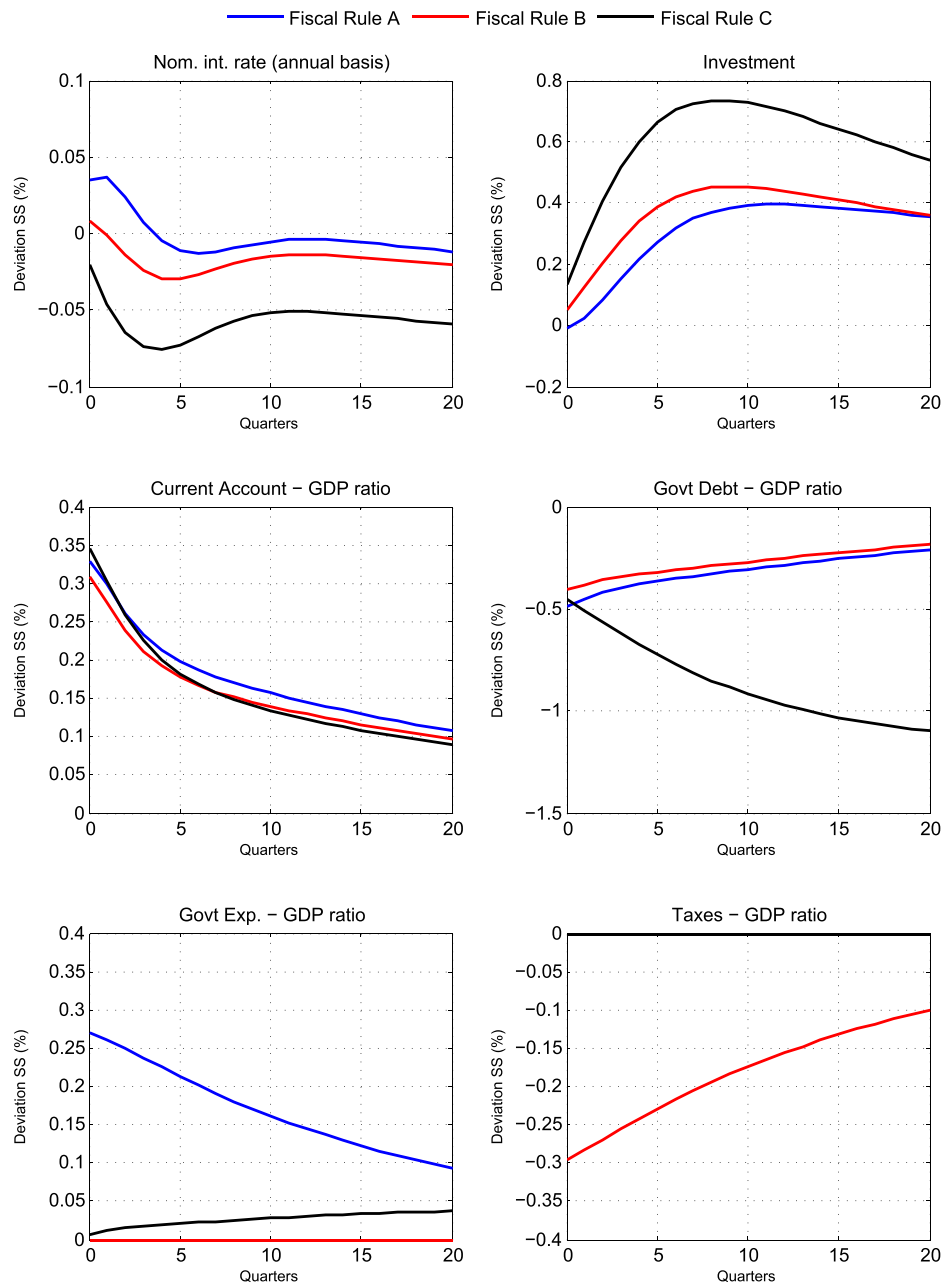


Fig. 3. Impulse-response to a copper price increase (cont.).

a natural question is whether the benefits of having the *structural balance* rule (or rule C) are still valid when there is a persistent drop in the copper price. Here we present the macroeconomic responses under the three fiscal rules when the economy experiences a big drop in the copper price.

To implement this, we modify the persistent coefficient for the copper price dynamics in order to capture highly permanent effects of shocks ($\rho_{cu} = 0.999$). Figs. 4 and 5 show the responses under the same three fiscal rules to a 50% persistent decline in the copper price. Although the fiscal rule C cannot avoid a contraction in GDP around 0.5% on impact, it can reduce the negative impact in comparison to rule A and B during the first two years after the shock. A similar pattern is observed for employment. The real exchange rate depreciates

substantially in response to this shock, but the magnitude of this effect is smaller under rule C. The rule C allows fiscal automatic stabilizers to operate. In fact, government expenditure rises as percentage of GDP in the first years after the shock. This allows a gradual transition of the economy to a persistent reduction in the copper price, avoiding an excessive cost of adjustment in the short-run. This evolution is similar to one described above on the recent developments in Chile. Thus, the gradual fiscal adjustment under the *structural balance* fiscal rule (rule C) after a big and persistent drop in the copper price reduces the short-run costs in the transition.

It is worth noting that financial position of the government is worsening to implement the adjustment gradually. Since the country premium elasticity is very low, this increase in the government

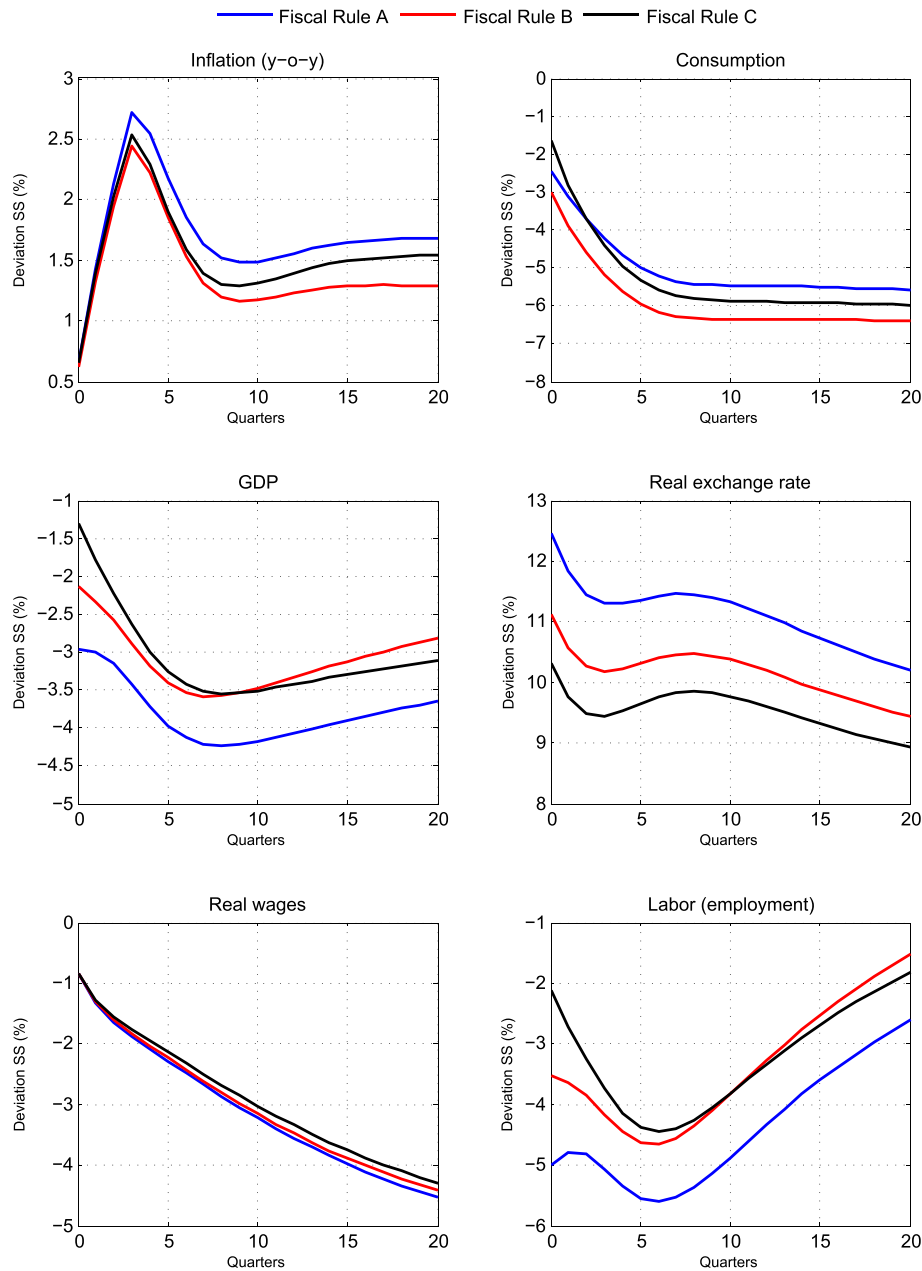


Fig. 4. Impulse-response to a persistent copper price drop.

debt implies a small rise in the external interest rate for the country. This case can be interpreted as a situation where the fiscal authority has enough space to increase debt or to reduce previous savings without affecting sovereign risk premiums. Nevertheless, this simulation might not be valid if the government has small savings or the level of government debt to GDP is high.

6.3. Interaction between fiscal and monetary policy

One of the puzzles in the recent business cycle in Chile is that despite the large increase in the copper price during 2005–2008 and 2010–2013, GDP did not grow as fast as in previous copper price booms. A direct explanation for this phenomena is the successful implementation of the *structural balance* fiscal rule that, as we saw

in the previous subsection, implies a muted response of government expenditure and GDP to a copper price shock.

However, there is an alternative explanation that is related in how monetary policy reacted previously to such shocks. As it was clear from the previous exercises, one of the effects of a copper price shock was to appreciate the real exchange rate. Over the 1990s, as the Chilean economy became more integrated to the world capital market and as rapid productivity growth attracted large capital inflows, there was a permanent tension between the needs for a tight monetary policy and concerns of an over appreciated exchange rate. Precisely, the restriction imposed by a target zone for the exchange rate limited the scope for an effective countercyclical monetary policy in response to shocks, such as the copper price shock we are analyzing.

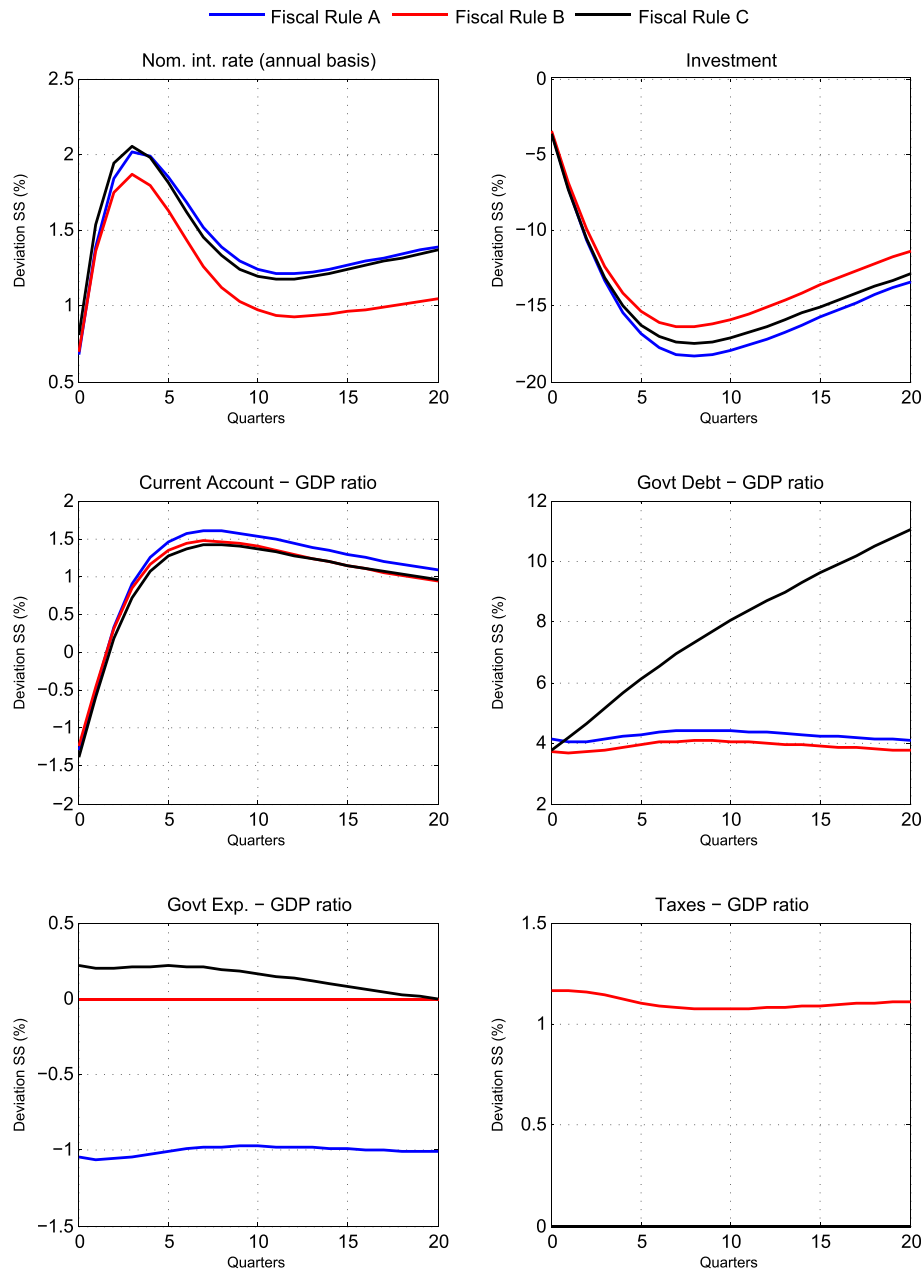


Fig. 5. Impulse-response to a persistent copper price drop (cont.).

To evaluate whether monetary policy constraints, derived from an implicit target for the real exchange rate, may have amplified business cycle response to copper price shocks, we compute impulse-responses assuming an alternative monetary policy that includes a response of the interest rate to real exchange rate deviations from trend. For this case we assume a coefficient for the real exchange stabilization of $\varphi_{rer} = 0.3$. A rule like the one proposed is an approximated representation of the behavior of the Central Bank of Chile over the 90s (see Caputo, 2005 and Schmidt-Hebbel and Tapia, 2002; Céspedes and Soto, 2007).

Figs. 6 and 7 present impulse-response functions to a 10% increase in the price of copper under fiscal rule A. From the figures it is clear that when the central bank tries to stabilize the real exchange rate in response to the shock, there is a larger response of output, consumption and employment. Notice that when the central bank follows this

policy, there is a decrease of real wages on impact. This is explained by the nominal wage rigidities and the larger increase of inflation under this monetary policy rule. Despite the fact that the response of output is in fact larger in this case, the magnitude of the effect produced by the attempt of the monetary authority in stabilizing the real exchange rate is not big. It would imply only a 0.2% extra expansion in GDP. It is worth noting that the looser monetary policy under the real exchange rate stabilization reduces significantly the real appreciation induced by the shock, and generates a big rise in inflation.

6.4. Imperfect credibility about the fiscal rule

As discussed before, the *structural balance* fiscal rule (rule C), by reducing the pro-cyclicality of the fiscal policy, isolates the economy

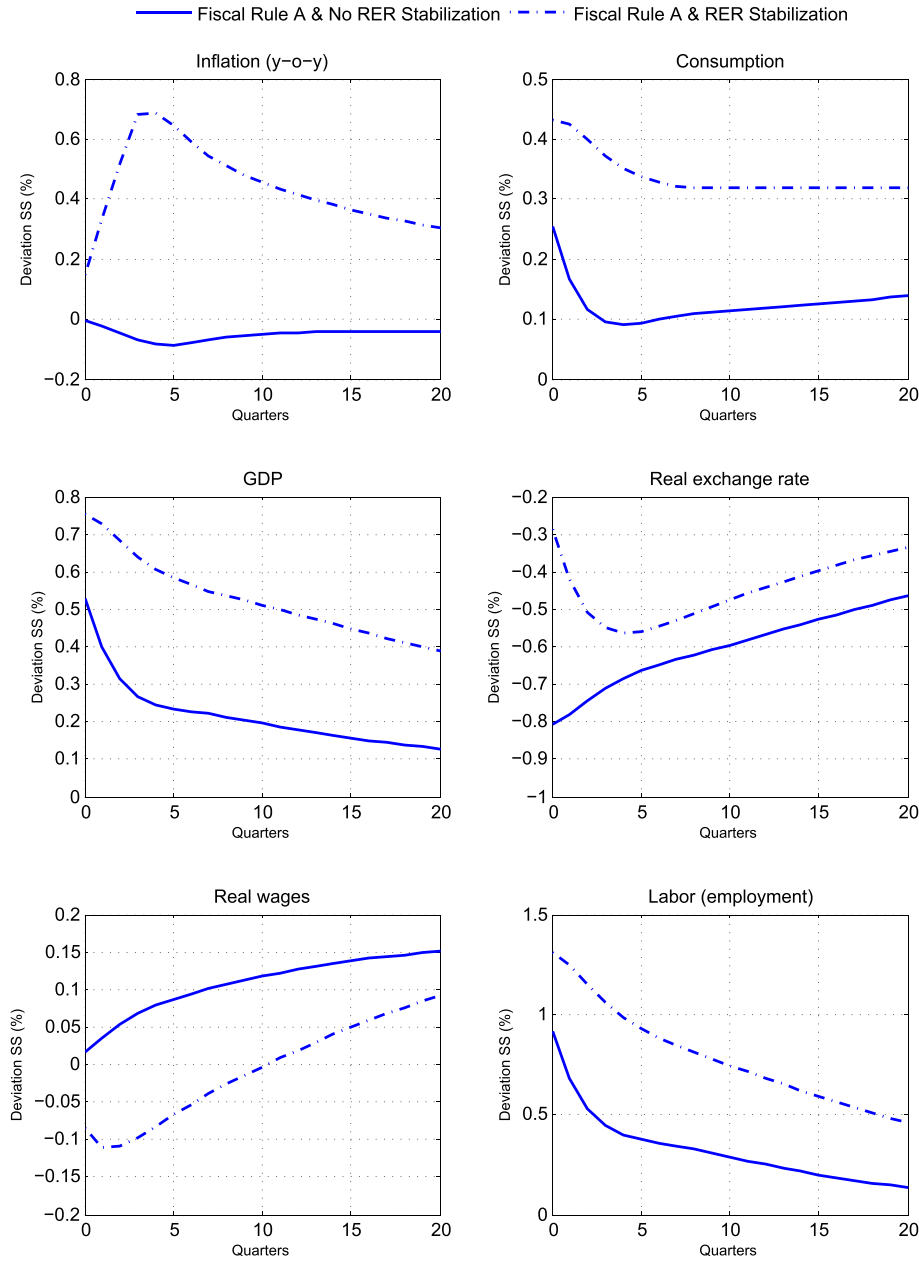


Fig. 6. Impulse-response to a copper price increase. Fiscal rule A and less flexible exchange rate.

from transitory shocks. Moreover, the fact that the rule is known to the private sector helps signaling the future behavior of the government. This affects current behavior of forward looking private agents. In this context, it is interesting analyzing in which way different degrees of credibility regarding this policy rule –i.e. different beliefs about the course of the fiscal policy in the future– affect the transmission mechanism of copper price shocks.

Under imperfect credibility, the private sector believes that, after a commodity price shock, government might rise spending by more than what would be prescribed by the rule C in Eq. (18). In practice, when there is an increase in the commodity price, fiscal revenues also rise. Then, the fiscal authority may be tempted to expand its expenditure beyond the rule C. Thus, we assume that under imperfect credibility, the private sector is uncertain if the fiscal authority

is type 1, following the rule (Eq. (18)) ; or it is type 2, in which case government consumption is given by

$$P_{G,t}G_t = T_t - \tilde{T}_t + \left(1 - \frac{1}{(1 + i_{t-1}^*)\Theta_{t-1}}\right) \varepsilon_t B_{G,t-1}^* - \bar{b}_S + \varphi_G \chi P_{S,t} Y_{S,t} \quad (21)$$

where φ_G determines the perceived degree of pro-cyclicality of fiscal policy in response to commodity price fluctuations of fiscal authority type 2.

Hence, we consider a case where fiscal authority actually behave according to their rule (Eq. (18)), but the private sector believes that

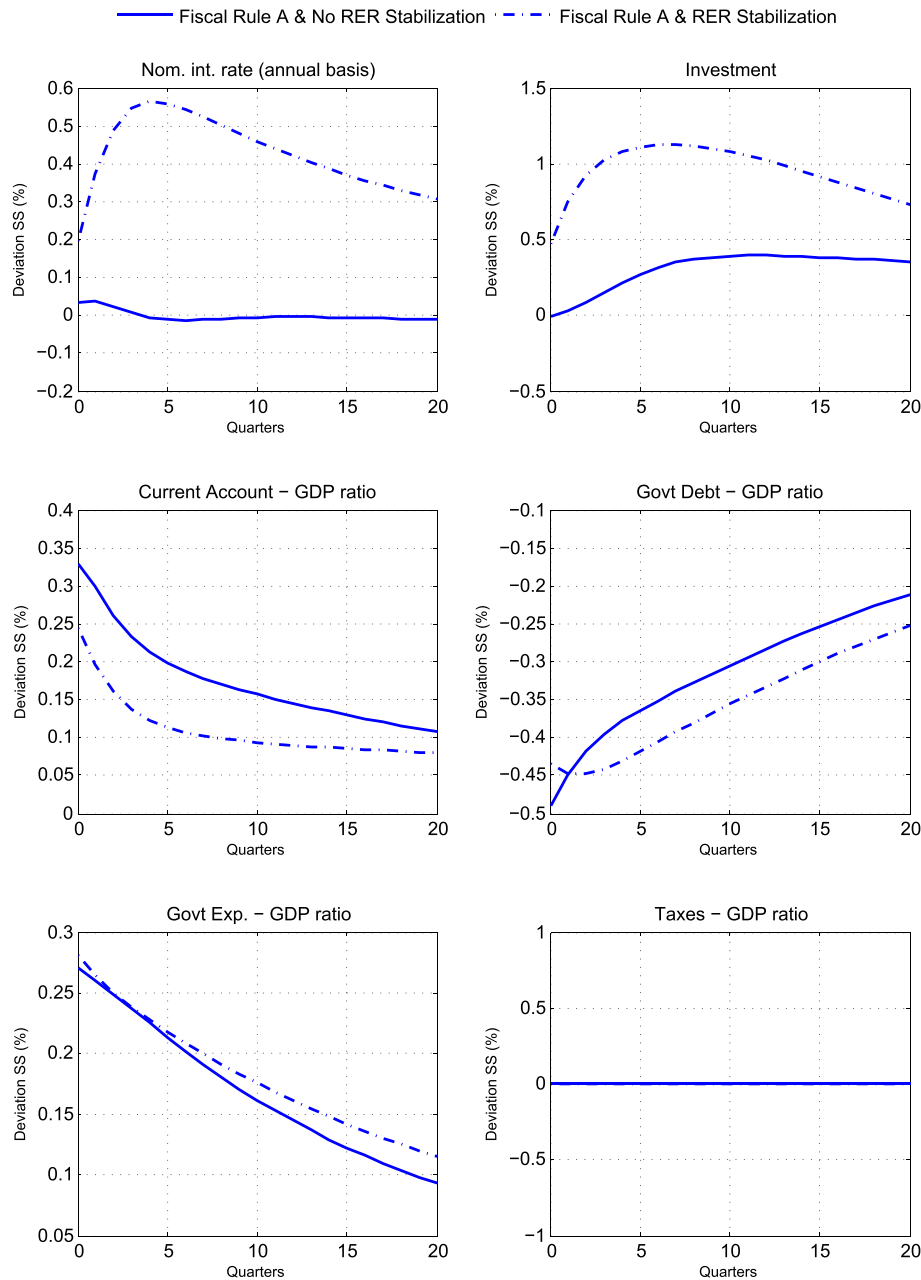


Fig. 7. Impulse-response to a copper price increase. Fiscal rule A and less flexible exchange rate (cont.).

with certain probability, fiscal authority is type 2 and follows rule (21). Notice that the possibility of shocks to the cyclical revenues, $\varepsilon_{T,t}$, makes it non-feasible for the private sector to infer the actual rule just by observing the behavior of fiscal policy and the other macroeconomic variables that enter in Eq. (18). It will be through a learning process that the private sector will eventually realize that macroeconomic policy instruments have not deviated from rule (18). Therefore, in order to infer the future path of the fiscal policy after a copper price shock, the private sector needs to solve a signal extraction problem. To that end, they use a Bayesian updating through the Kalman filter. To implement quantitatively this, we assume a standard deviation of shocks to cyclical revenues of 2% and $\varphi_G = 0.5$. We start the simulation assuming that the private sector assigns an equal prior probability for fiscal authority of being type 1 and 2.

The responses to a 10% increase in copper price under the rule C for perfect and imperfect credibility is shown in Figs. 8 and 9.

A copper price shock leads to a smaller decrease in inflation under imperfect credibility compared to the case of perfect credibility. Also the real appreciation that follows the shock is stronger under imperfect credibility. Also, the response of output is higher when the private sector is uncertain about the behavior of the fiscal authority in response to the copper price increase. Also, consumption rises by more under imperfect credibility. Interestingly, investment increases by less under imperfect credibility because the path for the monetary policy rate is higher under imperfect credibility. Thus, imperfect credibility about the *structural balance* fiscal rule amplify mainly the effects of copper price shocks on GDP, employment, and the real exchange appreciation.

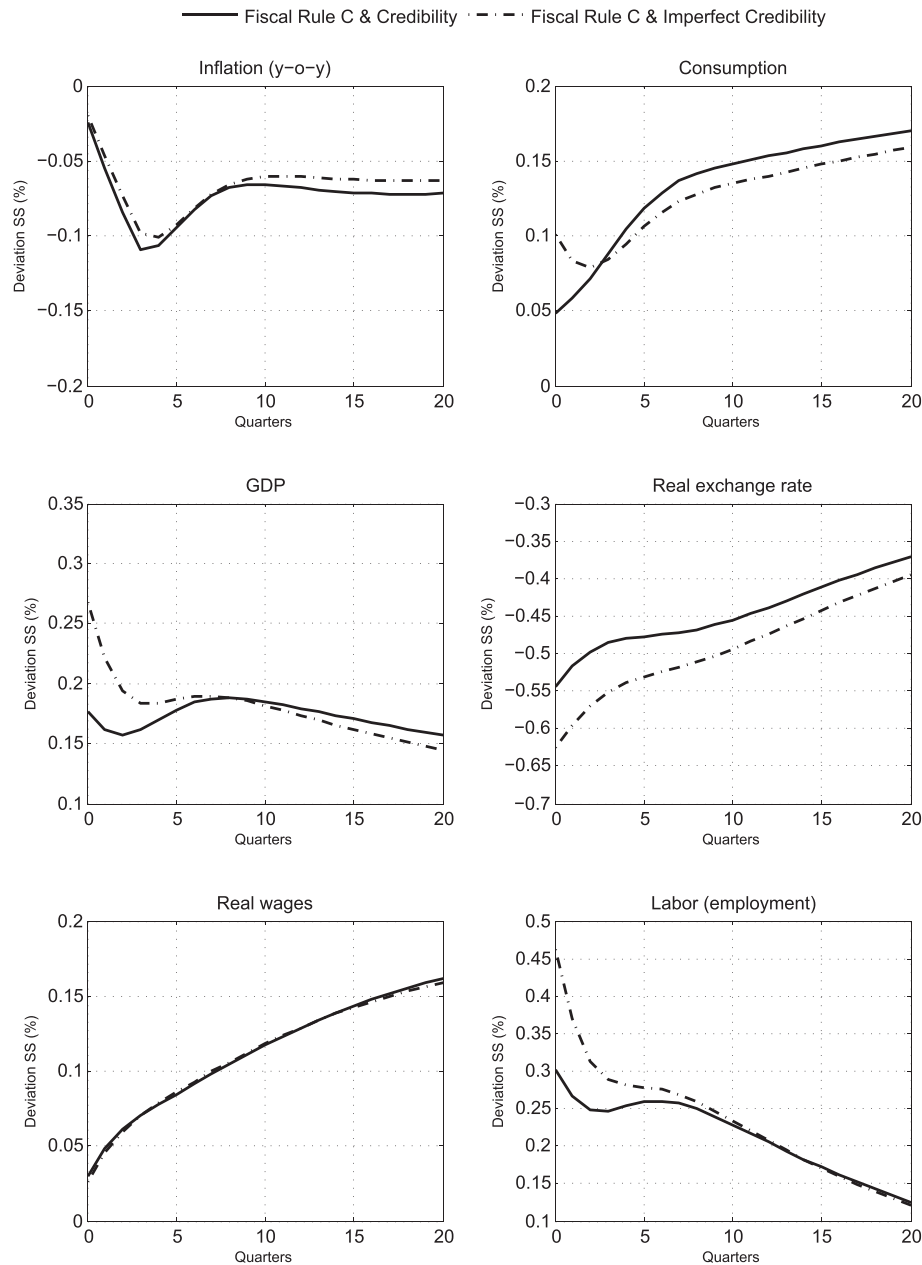


Fig. 8. Impulse-response to a copper price increase. Imperfect credibility.

7. Conclusions

In this paper we use a dynamic stochastic general equilibrium (DSGE) model for the Chilean economy to analyze the effect of copper price shocks. We compare the results of copper price shocks under different regimes for fiscal and monetary policies. The results show that when the fiscal policy is highly expansive in response to a transitory copper price shock, and if an important fraction of households are non-Ricardian, then an increase in the copper price of 10% implies an output expansion of almost 0.6%. The real exchange rate appreciates by 0.8%. In contrast, if the fiscal policy is conducted according to the *structural balance* rule (rule C), then output would increase much less, around 0.1–0.2%, and the real exchange rate would appreciate by 0.5% in response to the same copper price increase. When there is a persistent drop in copper price, our results are still valid. Thus, the *structural balance* rule also isolates better the

economy after a persistent decline in the copper price, providing a gradual fiscal adjustment and reducing the macroeconomic costs in the short-run.

Using the model we analyze the implications of exchange rate flexibility. Our results indicate that the adoption of a flexible exchange rate regime has also contributed to reduce the vulnerability of the economy to copper price fluctuations.

Our analysis also highlights that imperfect credibility about the *structural balance* fiscal rule tends to amplify the effects of the copper price shocks on GDP, employment and the real exchange rate. However, the effects of this type of shocks on investment are reduced under imperfect credibility. This reflects dependence of investment decisions on the future expected path of monetary policy. Since under imperfect credibility the expected path of the monetary policy rate is higher after a copper price increase, investment increases by less when the *structural balance* fiscal rule is imperfectly credible.

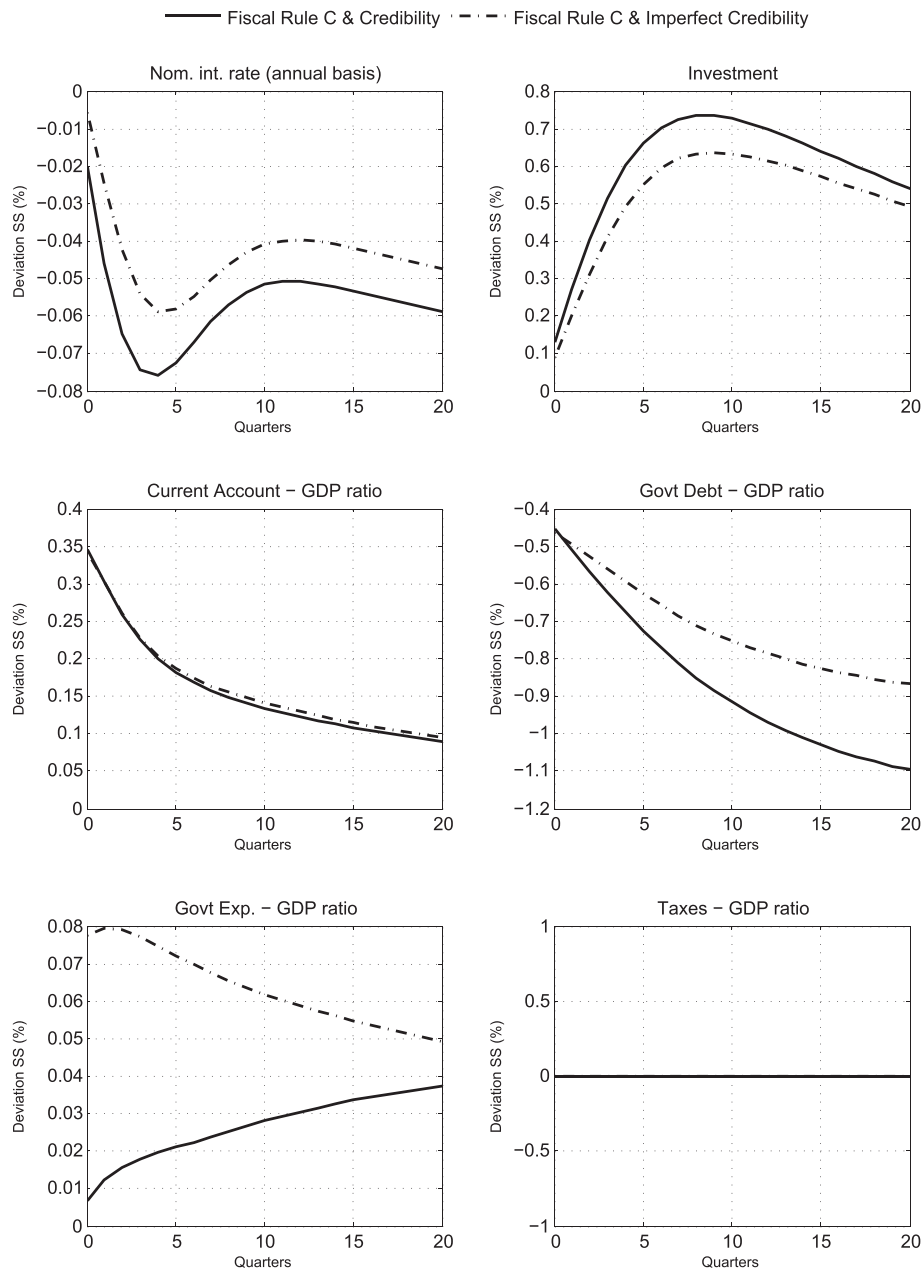


Fig. 9. Impulse-response to a copper price increase. Imperfect credibility (cont.).

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